

1. Use function `random()` from the `random` Python module to estimate the value of π . Your algorithm must be based on the given random generator (you may not use any alternative algorithm for generating the value of π) and you may not use any other package/library.
2. Covariance matrix for a multivariate random variable is defined as $\Sigma_{xx} = E((x - \mu_x)(x - \mu_x)^T)$. The cross covariance matrix for two multivariate random variables is defined as $K_{xy} = E((x - \mu_x)(y - \mu_y)^T)$.
 - a. Express the covariance of a sum of two multivariate random variables in terms of covariance and cross-covariance.
 - b. Prove that the cross-covariance of two independent multivariate random variables is a zero-matrix.
 - c. What is the corollary of the above two results?
3. A robot has 10 (unreliable) sensors that independently detect the presence of an object in its intake mechanism. The probability of detecting the object for each sensor is 0.1.
 - a. Write the distribution of the random variable that represents how many sensors have detected the presence of an object?
 - b. What is the probability that at least one sensor has detected the presence of an object?
4. A two-dimensional normally-distributed random variable has the uncertainty ellipse centered at (3, 4) and one principal axis of the ellipse forms a 30-degree angle with the x-axis of the coordinate system. The variance associated with that axis is 1. The variance associated with the other axis is 0.25. Write the probability density function of this random variable.
5. Suppose there is a village in which every family wants to have a boy. If they get a boy, they stop having more children, if they get a girl, they continue to have more children. What is the average number of children per family?

1. Use function `random()` from the `random` Python module to estimate the value of π . Your algorithm must be based on the given random generator (you may not use any alternative algorithm for generating the value of π) and you may not use any other package/library.

python `random.random()` takes rand. number $[0,1)$ (e.g, 0.1 or 0.7)

Apply this function twice, $x_{rand} = rand()$ & $y_{rand} = rand()$, you get a 2d dimensional coordinate within a 1×1 square where $[0,1)$.

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{x^2 + y^2} \leq r, \text{ where } r = 1$$

$$\sqrt{x^2 + y^2}^2 \leq 1^2 \iff (x^2 + y^2) \leq 1$$

Taking known axiom that a square area is $2\pi r^2$, we want to take for all events Ω , $p(\Omega) = 1$, we want all events A where $\sqrt{x^2 + y^2} \leq r$ or, in other words points that lie within the $r=1$ radius. desired to be ratio $p = \pi r^2 / 1$, so in code for all M point $\in N$ events where expected point ≤ 1 ,

```

N = 10000
M = 0
for range(N):
    if (rand()2 + rand()2) <= 1 :
        M += 1
ratio = M / N
pi = 4 * ratio

```

also in `hw-1.py`

2. Covariance matrix for a multivariate random variable is defined as $\Sigma_{xx} = E((x - \mu_x)(x - \mu_x)^T)$. The cross covariance matrix for two multivariate random variables is defined as $K_{xy} = E((x - \mu_x)(y - \mu_y)^T)$.
- Express the covariance of a sum of two multivariate random variables in terms of covariance and cross-covariance.
 - Prove that the cross-covariance of two independent multivariate random variables is a zero-matrix.
 - What is the corollary of the above two results?

Let x and y be random vectors in the same dimension n

(a) covariance of a sum of two multivariate rand variable in terms of covariance & cross covariance. This can be written as $\Sigma_{(x+y), (x+y)}$

substituting this into the covariance formula:

$$\Sigma_{(x+y), (x+y)} = E((x+y - \mu_x + \mu_y)(x+y - \mu_x + \mu_y)^T).$$

using the linearity of expectation $E[a+b] = E[a] + E[b]$

$$\Sigma_{(x+y), (x+y)} = E(((x - \mu_x) + (y - \mu_y))(x - \mu_x) + (y - \mu_y)^T)$$

taking $a = (x - \mu_x)$ & $b = (y - \mu_y)$, we can write the above as $E((a+b)(a+b)^T)$. Now, expand

$$E(aa^T + ab^T + ba^T + bb^T)$$

$$= E((x - \mu_x)(x - \mu_x)^T) + E((x - \mu_x)(y - \mu_y)^T) + E((y - \mu_y)(x - \mu_x)^T) + E((y - \mu_y)(y - \mu_y)^T)$$

$$= \Sigma_{xx} + K_{xy} + K_{yx} + \Sigma_{yy}$$

Since $E(xy^T) = E(yx^T)^T$

$$= \Sigma_{xx} + K_{xy} + K_{xy}^T + \Sigma_{yy}$$

(b) prove $K_{xy} = 0$ when x, y are independent.

Recall $K_{xy} = E((x - \mu_x)(y - \mu_y)^T)$

using independent expectation, $E(uv) = E(u) \cdot E(v)$

we can rewrite as $K_{xy} = E(x - \mu_x) \cdot E(y - \mu_y)$

Now, the centered variable mean $E(x - \mu_x)$ can be written as $E(x) - E(\mu_x)$ and since μ_x is uniform, $E(x) - E(\mu_x) = \mu_x - \mu_x = 0$

Since $K_{xy} = E(x - \mu_x) \cdot E(y - \mu_y)$ and $E(x - \mu_x) = 0, E(y - \mu_y) = 0$

$K_{xy} = 0$

(c) Combining the 2: $\Sigma_{(x+y), (x+y)} = \Sigma_{xx} + K_{xy} + K_{xy}^T + \Sigma_{yy}$ & $K_{xy} = 0$ if $x \perp y$

when $x \perp y \Rightarrow \Sigma_{(x+y), (x+y)} = \Sigma_{xx} + \Sigma_{yy}$

3. A robot has 10 (unreliable) sensors that independently detect the presence of an object in its intake mechanism. The probability of detecting the object for each sensor is 0.1.
- Write the distribution of the random variable that represents how many sensors have detected the presence of an object?
 - What is the probability that at least one sensor has detected the presence of an object?

a. for 10 sensors (events), we can write the following

$$p \cdot p \cdot p \dots (1-p) \dots (1-p) = p^k \cdot (1-p)^{n-k}$$

$$X \sim \text{Binomial}(n=10, p=0.1)$$

$$P(X=k) = \binom{10}{k} 0.1^k \cdot 0.9^{10-k}, \quad k = 0 \sim 10$$

where $E[X] = np$ or 1

(b) at least one detected, in other words from all event subtract the times no detector fired.

$$P(X=0) = \binom{10}{0} 0.1^0 \cdot 0.9^{10} = 0.9^{10} \approx 0.3486$$

so $P(X \geq 1) = 1 - P(X=0) \approx 0.6514 \%$

4. A two-dimensional normally-distributed random variable has the uncertainty ellipse centered at (3, 4) and one principal axis of the ellipse forms a 30-degree angle with the x-axis of the coordinate system. The variance associated with that axis is 1. The variance associated with the other axis is 0.25. Write the probability density function of this random variable.

mean $\mu = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

Recall multivariate PDP: $P(x|\mu, \Sigma) = \frac{1}{2\pi^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$

Building variance Σ from axis

$$\Sigma = R \Lambda R^T, \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \text{Rotation Matrix } R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

plug in numbers:

$$\Sigma = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0.25 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\Lambda R^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \quad R \Lambda R^T = \begin{bmatrix} \frac{3}{4} + \frac{1}{16} & \frac{\sqrt{3}}{4} - \frac{3}{16} \\ \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{16} & \frac{1}{4} + \frac{3}{16} \end{bmatrix} = \begin{bmatrix} \frac{13}{16} & \frac{3\sqrt{3}}{16} \\ \frac{2\sqrt{3}}{16} & \frac{7}{16} \end{bmatrix}$$

$$\Sigma = \frac{1}{16} \begin{bmatrix} 13 & 3\sqrt{3} \\ 3\sqrt{3} & 7 \end{bmatrix}$$

$$\det \Sigma = \lambda_1 \lambda_2 = 0.25 = \frac{1}{4}$$

solving for $\frac{1}{2\pi^{d/2} |\Sigma|^{1/2}} = \frac{1}{2\pi \sqrt{\frac{1}{4}}} = \frac{2}{2\pi} = \frac{1}{\pi}$

solving for $\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$

$$\Sigma^{-1} = \frac{1}{\det} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{\frac{1}{4}} \frac{1}{16} \begin{pmatrix} 7 & -3\sqrt{3} \\ 3\sqrt{3} & 13 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 7 & -3\sqrt{3} \\ -3\sqrt{3} & 13 \end{pmatrix}$$

$$u = x - 3 \quad v = y - 4 \quad \text{and } (x - \mu)^T = (u \ v) \quad \text{and } (x - \mu) = \begin{pmatrix} u \\ v \end{pmatrix}$$

$$(u, v) \Sigma^{-1} \begin{pmatrix} u \\ v \end{pmatrix} = 6'' u^2 + 26'' uv + 6'' v^2 = \frac{7}{4} u^2 - \frac{6\sqrt{3}}{4} uv + \frac{13}{4} v^2$$

$$-\frac{1}{2} (u, v) \Sigma^{-1} \begin{pmatrix} u \\ v \end{pmatrix} = -\frac{1}{8} (7u^2 - 6\sqrt{3} uv + 13v^2)$$

$$\frac{1}{2\pi^{\frac{n}{2}}|\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

$$= \frac{1}{\pi} \exp\left(-\frac{1}{8}(7u^2 - 6\sqrt{3}uv + 13v^2)\right)$$

$$= \frac{1}{\pi} \exp\left(-\frac{1}{8}\left(7(x-3)^2 - 6\sqrt{3}(x-3)(y-4) + 13(y-4)^2\right)\right)$$

or in compact matrix form

$$\mu = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \Sigma = \frac{1}{16} \begin{pmatrix} 13 & 3\sqrt{3} \\ 3\sqrt{3} & 7 \end{pmatrix}, \quad f(x) = \frac{1}{2\pi \det \Sigma^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

5. Suppose there is a village in which every family wants to have a boy. If they get a boy, they stop having more children, if they get a girl, they continue to have more children. What is the average number of children per family?

policy family will always have 1+ child ($x \geq 1$)
last child will be boy $\sum_i^n$, where $P(X=n+1) = 0.5 = \frac{1}{2}$

$$E = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2} E \Rightarrow E = 2$$

Every family has at least 1 child. $P(x) = \frac{1}{2}$ time its a boy and the family stops. The other half its a girl and the family is exactly in the same spot (ind. every turn). Since the chance of boy or girl reset every time, we can write $\frac{1}{2} \cdot E$